

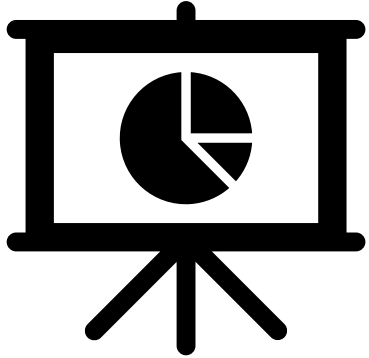
Activity 05: Refinement

SISMID Lab

July 22 through July 24, 2026

A Human – AI Teaming Approach to Infectious Disease Modeling

General Housekeeping

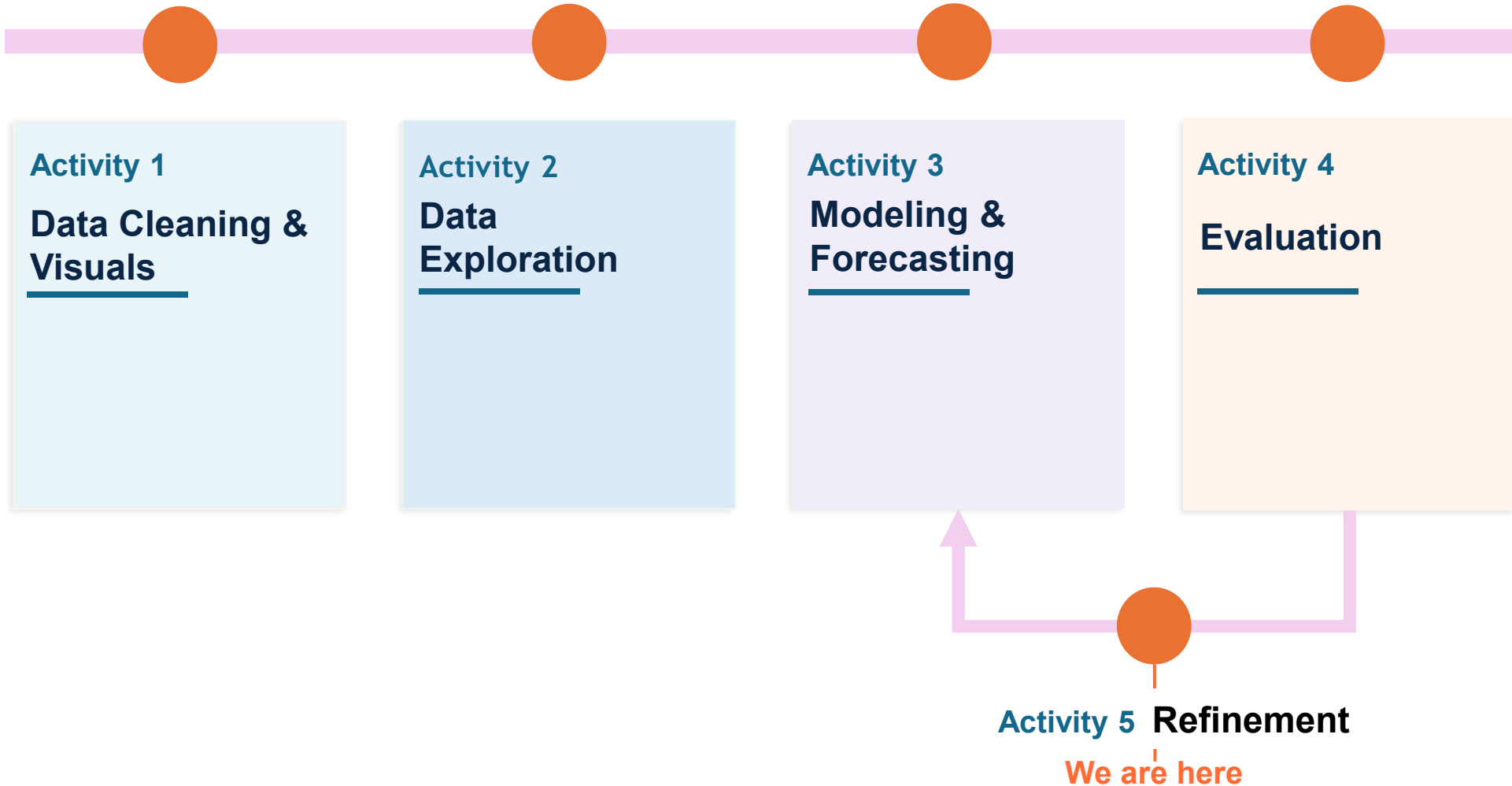


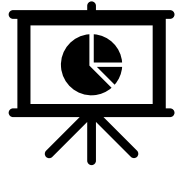
Any slide with this icon contains the **overview** and **expected outputs** for the included tasks. All outputs listed on these slides should be uploaded to the course GitHub repository for review.



Any slide with this icon indicates that it is **time to complete a hands-on activity**. Carefully follow the instructions provided, complete the required task, and verify your results.

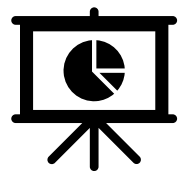
Lab Timeline





Task Overviews

Activity	Description
Task 1: Leading Indicators	We will walk through how to incorporate additional data sources into our model using AI. <u>You will choose</u> the source and variable and decide whether to keep it or drop it from the model.
Task 2: Model Type	We will walk through some additional forecasting model options and how to switch our forecasting model using AI. <u>You will choose</u> if you want to switch to another model type.
Task 3: Free Coding & FluSight Challenge	We will quickly walk through some set-up rules for FluSight submissions. You will then have some time to free code and work on improving your model. Using your final model, you can submit it to our mini-challenge to see how your model compares to others in the course.



Goal Output

From the three tasks (see previous slide), we should get the below-listed output. By the end of this activity, you should have a completed **FluSight model** as well as the **full pipeline** (including the script for this activity; see below) used to build and implement the model.

Activity Output	Description
<i>rules.md</i>	Forecasting rules for Tasks 1 through 3 — leading indicators, changing the model, “FluSight” submission
<i>agents.md</i>	Instructions for the AI assistant (the "agent"). It tells the AI who it is, what job to do, and what rules to follow. You update this file to change how the AI behaves.
<i>scripts/05_improvements.R</i>	R Script that houses all our model changes. The goal is to not overwrite our previous changes, rather try out new things.
<i>YY-MM-DD_NAME_MODEL.csv</i>	Collection of FluSight submission files. We will cover how to achieve these, and the final products will be uploaded to GitHub.

Modeling Process

- The modeling process is iterative; you are **not** going to get it perfect right away.
- Our ARIMA forecasts are **nowhere near perfect** at capturing the truth, even the adjusted non-perfect truth.
- We can try different things to improve model performance, using AI to help us implement the changes:

Leading Indicators

An outside signal that rises before admissions do.

Different Models

A different structure:
GAM, XGBoost,
ensembles, etc.

Task One

Leading Indicators

Leading Indicators

- **A leading indicator moves *before* the thing you're forecasting.** It climbs or dips weeks ahead of hospital admissions, carrying an early-warning signal the admissions series alone doesn't hold.
- **Why it can help our models:** right now, the model sees only past admissions. A leading indicator feeds it outside information that may anticipate turns the admissions history hasn't shown yet.
- **Same forecast, more context.** We hold everything else fixed and ask one narrow question: does this single extra signal make next week's forecast better?

Please Note: When selecting leading indicators, keep the disease process in mind. What naturally comes *before* a person gets hospitalized? We want our model to **perform well** while **remaining explainable**.

Other Available Data Sources

CDC National Syndromic Surveillance Program

NSSP Emergency Department Visit Trajectories by State and Sub-State Regions (COVID-19, Flu, RSV, Combined). Reports the weekly percentage of emergency department visits attributable to each pathogen, out of all ED visits, for a given state or Health Service Area (HSA). HSAs are groups of one or more counties that align to patterns of care seeking, allowing trends to be viewed below the state level.

Other Available Data Sources

CDC National Wastewater Surveillance System

NWSS Wastewater Viral Activity Levels by National, State, Region, and Site (Influenza A). Reports the concentration of pathogen genetic material detected in untreated wastewater, expressed as a viral activity level relative to a site's historical range and updated weekly. Each site covers a sewershed, the community area served by a wastewater collection.

Examples: Leading Indicators

CDC (NHSN) – Inpatient Admissions	CDC (NSSP) – ED Visits*	CDC (NWSS) – Wastewater**
Total Pediatric Influenza Admissions	3-week moving average of % ED Visits with a discharge diagnosis of Influenza (percent_visits_smoothed_influenza)	National Wastewater Viral Activity (National WVAL)
Percent Change in the Total Influenza Admissions from Prior Week	% ED Visits with a discharge diagnosis of Influenza (percent_visits_influenza)	–
Number of Adult Influenza Admissions, 18–49 years	–	–

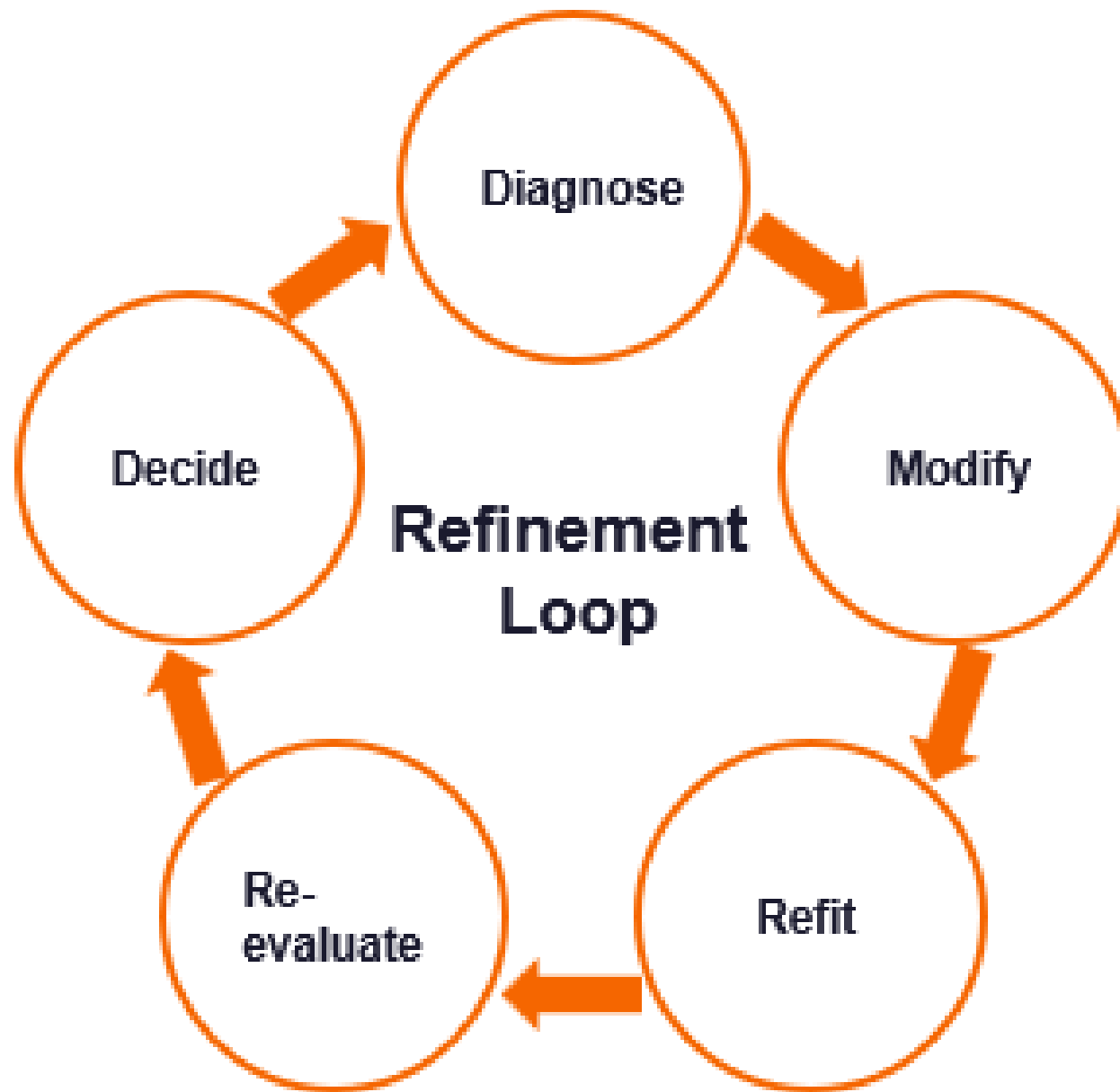
* For additional information on the CDC(NSSP) data set, please see <https://cmu-delphi.github.io/delphi-epidata/api/covidcast-signals/nssp.html>

** For additional information on the CDC(NWSS) data set, please see <https://www.cdc.gov/wastewater/about/data-methods.html>

Model Refinement: An Iterative Loop

In infectious disease forecasting, refinement means improving the forecasting system based on what we learn from evaluation.

- 1. Diagnose** — Where does performance break down? By horizon, season phase, geography, outcome, or uncertainty.
- 2. Modify** — Make a targeted change to the data, predictors, model structure, or uncertainty estimates.
- 3. Refit** — Train the revised model using the same forecasting setup.
- 4. Re-evaluate** — Compare forecasts against held-out observations.
- 5. Decide** — Keep the refinement only if it improves out-of-sample forecast performance (i.e., in testing period).



**Just as a reminder: We are using the
FluSight format for all forecasts.**

Column	Description	Entry Type	Acceptable Values
reference_date	The last day of the week the forecast(s) are submitted to the challenge. It is the Saturday following the forecast due date.	ISO Format Date	YYYY-MM-DD
target	The target of the forecast.	Character	wk inc flu hosp
horizon	The number of weeks between the reference_date and the target_end_date.	Numeric	[1]
target_end_date	The last date of the retrospective/nowcast/forecast target's week (Saturday).	ISO Format Date	YYYY-MM-DD
location	FIPS codes for U.S. states as well as "US" for national forecasts the corresponding row is associated with.	Character	US
output_type	The type of value shown.	Character	quantile
output_type_id	The quantile label for the value in the corresponding row.	Numeric	0.01, 0.025, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 0.975, and 0.99
value	The non-negative numbers indicating the quantile prediction for the corresponding row.	Numeric	Integer

Evaluation: Should we keep the change?

We measure each change against the current model using our Activity 4 evaluation script (**Hint:** we will include that in our rules to AI for this activity), then decide whether to keep it or revert.

Change	Model	Indicator	WIS	MAE	95% PI Coverage	Keep?
Baseline	ARIMA	N/A	4862.8	6102.2	85%	—
Leading Indicator	ARIMA	National WVAL	[]	[]	[]	[Y / N]

Please Note: The metrics shown in the table correspond to the mean metrics at the 3-week horizon. We have selected the wastewater data for this example; feel free to select another option. You will also have time at the end to explore more.



Now it's your turn!

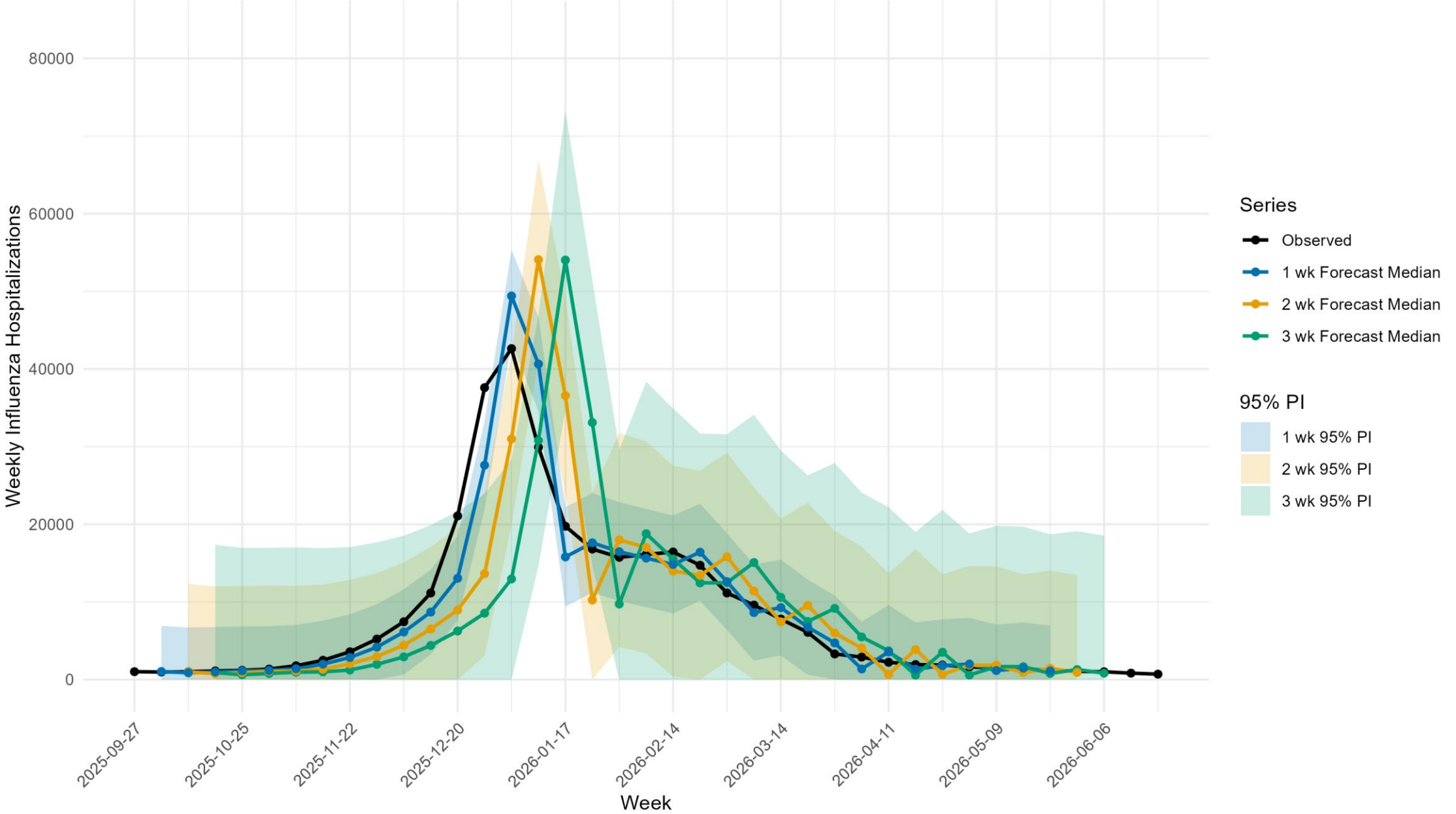
We will work on incorporating one leading indicator **together**, and there will be time for you to explore a bit more later in the module.

1 Add the `## Leading Indicators` header (either prompting AI to add or manually doing it).

We will then prompt Copilot (or AI of your choosing) to incorporate this in the pipeline (i.e., rules, agent, script), while **maintaining what we had done before**. Rules should include **preparing the forecast** in the FluSight format (**Hint:** Activity 3) and producing evaluation results (**Hint:** Activity 4). Dependent on the prompt of the AI model used, we may get different results. Therefore, it's important we check our results to make sure the AI suggestion is methodically valid.

3 After updating the **rules.md**, **agents.md**, and creating the **script**, run it and check the results.

USA 1-, 2-, & 3-Week-Ahead Influenza Hospitalization Forecast, Wastewater-Informed ARIMAX (2025-26 Season)



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Change	Model	Indicator	WIS	MAE	95% PI Coverage	Keep?
Baseline	ARIMA	N/A	4862.8	6102.2	85%	—
Leading Indicator	ARIMA	National WVAL	4271.1	5407.0	91%	[Y / N]

Please Note: The metrics shown in the table correspond to the mean metrics at the 3-week horizon. We have selected the wastewater data for this example; feel free to select another option. You will also have time at the end to explore more.

Task Two

Changing the Model Type

Meet XGBoost

XGBoost builds a rough model, sees where it is wrong, and then builds another model to fix those errors. It repeats this a few hundred times, and that process is what we call gradient boosting.

What it is

Short for Extreme Gradient Boosting. A **single decision tree** is like a game of 20 Questions: it asks simple yes/no questions about the data until it reaches an answer. XGBoost **teams up many small trees**, each fixing the last one's mistakes, to make one accurate prediction.

How it learns

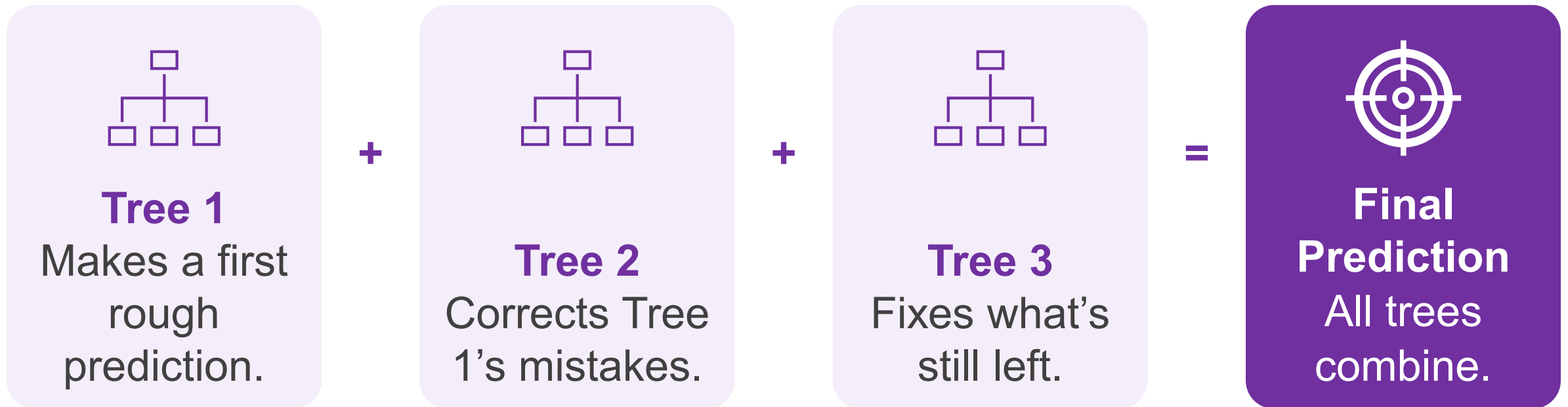
Each new tree **targets** what the **current model** still gets **wrong**, and only a fraction of its correction is applied. Learning slowly and in sequence is what keeps the model from **overfitting** to noise.

Why it shines

It captures complex, **nonlinear patterns** and **feature interactions** on its own, copes well with **many predictors** and **missing values**, and is consistently a top performer on structured, tabular data.

How XGBoost Works

Boosting adds trees one at a time — each new tree fixes the errors the ones before it left behind.



More trees, sharper predictions — the leftover errors keep shrinking.

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Baseline	ARIMA	N/A	4862.8	6102.2	85%	—
Leading Indicator	ARIMA	National WVAL	4271.1	5407.0	91%	[Y / N]
Model Swap	XGBoost	N/A	[]	[]	[]	[Y / N]

Please Note: The metrics shown in the table correspond to the mean metrics at the 3-week horizon. We have selected the XGBoost model for this example; feel free to select another option. You will also have time at the end to explore more.



Now it's your turn!

We will work changing the model **together**, and there will be time for you to explore a bit more later in the module.

1

Add the `# Model Improvements` header, and subsequent `## Model Change` sub header (either prompting AI to add or manually doing it).

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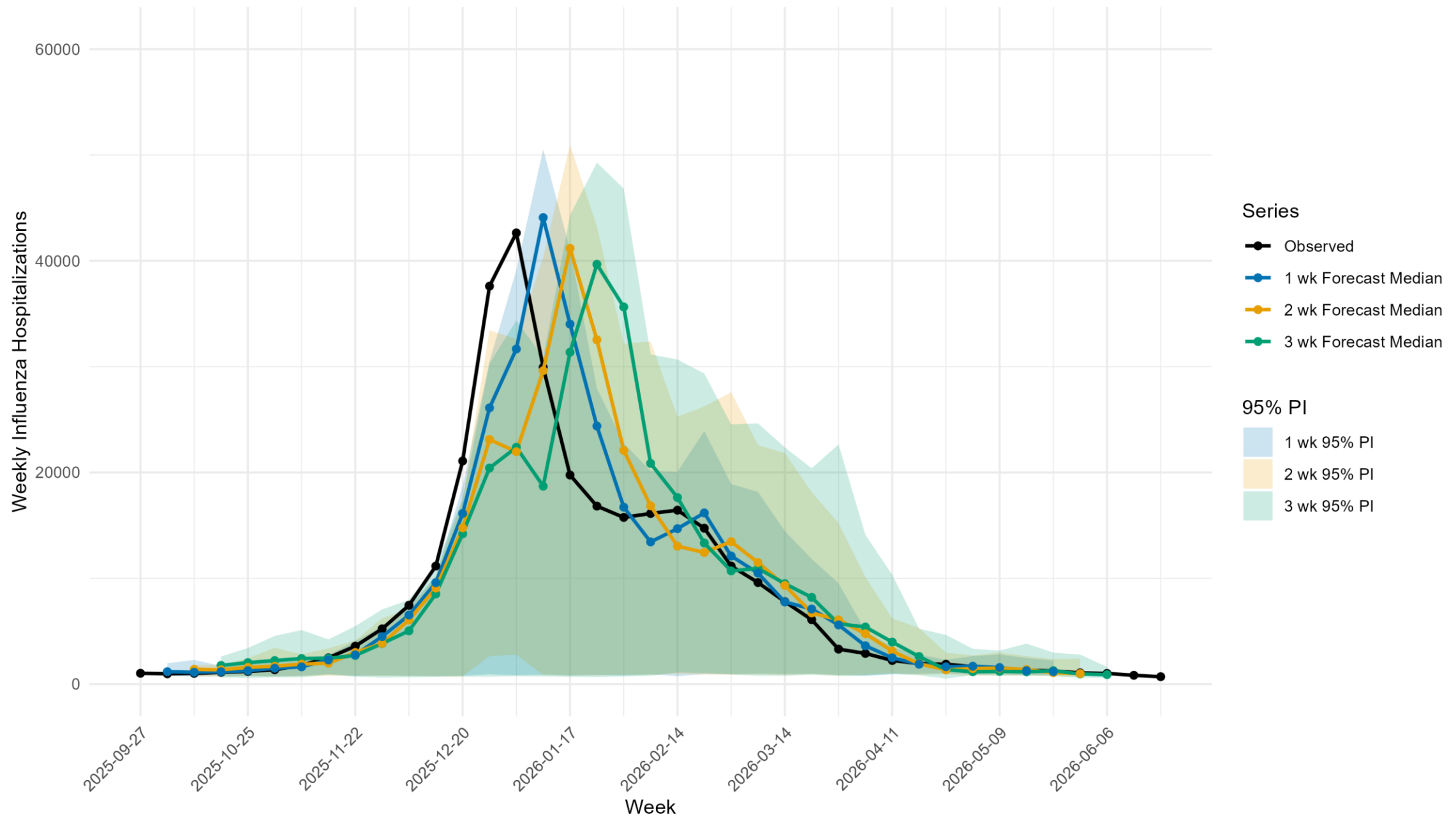
2

Rules should include **preparing the forecast** in the FluSight format (**Hint:** Activity 3) and producing evaluation results (**Hint:** Activity 4). Dependent on the prompt of the AI model used, we may get different results. Therefore, its important we check our results to make sure the AI suggestion is methodically valid.

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After updating the **rules.md**, **agents.md**, and creating the **script**, run it and check the results.

USA 1-, 2-, & 3-Week-Ahead Influenza Hospitalization Forecast, XGBoost Quantile Model (2025-26 Season)



Evaluation: Should we keep the change?

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Baseline	ARIMA	N/A	4862.8	6102.2	85%	—
Leading Indicator	ARIMA	National WVAL	4271.1	5407.0	91%	[Y / N]
Model Swap	XGBoost	N/A	2634.8	4179	91%	[Y / N]

Please Note: The metrics shown in the table correspond to the mean metrics at the 3-week horizon. We have selected the XGBoost model for this example; feel free to select another option. You will also have time at the end to explore more.

Task 3: Free Coding & Mini-Challenge!

<https://github.com/bleicham/Human-AI-Forecasting-Challenge-Sandbox>